Groupoid or binary algebra: Definition
A non empty set G equipped with one binary operation '*' is called groupoid.
i.e. G is a groupoid if G is closed for *.
It is denoted by $(G, *)$.
for example: $(N, +)$, $(Z, -)$, $(Q, \times)$ etc.
Note: Groupoid is also called Quasi Group

$$N = \{1, 2, 3, 4 \ldots\}$$

$$Z = \{\ldots -3, -2, -1, 0, 1, 2 \ldots\}$$

$-2 \& 3 \in Z$

$-2+3 \in Z$
$-2\times 3 \in Z$
$-2-3 \in Z$
$-2 \div 3 \notin Z$

if any Group that satisfy the condition of being closed it is called groupoid otherwise it is called Quasi Group.

A closed group is a group that belongs its own group for example in this case after performing +, −, and multiplication operations on −2 and 3 the result is an integer (groupoid) but after division the result is not integer hence a Quasi group.

SemiGroup: Definition
An algebraic structure $(G, *)$ is called group if the binary operation * satisfies associative property i.e.
$[G_1]$ $(a*b)*c = a*(b*c)$, $\forall\ a \in G$

Ex. 1 The algebraic structures $(\mathbb{N},+)$, $(\mathbb{Z},+)$, $(\mathbb{Z},\times)$, $(\mathbb{Q},\times)$ are semigroups but the structure $(\mathbb{Z},-)$ is not sobecause subtraction $(-)$ is not associative.

Ex. 2 The structures $(P(S),\cup)$ and $(P(S),\cap)$ where $P(S)$ is the powerset of a set S are semigroups as both the operations union $(\cup)$ and intersection $(\cap)$ are associative.

$$\mathbb{N}=\{1,2,3,\dots\}$$

$2,4,7\in\mathbb{N}$

$2+(4+7)=(2+4)+7$

or $2\times(4\times7)=(2\times4)\times7$

and $2-(4-7)\neq(2-4)-7$

Monoid: Definition

A semigroup is called monoid if there exists an identify element 'e' in G such that:

[G2] $e*a=a*e=a,\ \forall\ a\in G$

Ex. 1. The semigroup $(\mathbb{N},\times)$ is a monoid because 1 is the identity for the multiplication. But the semigroup $(\mathbb{N},+)$ is not, because 0, is the identity for addition, is not in $\mathbb{N}$.

Ex. 2. The semigroups $(P(S),\cup)$ and $(P(S),\cap)$ are monoid because $\emptyset$ and S are the identities respectively for union $(\cup)$ and intersection $(\cap)$ in $P(S)$.

$N = \{1, 2, 3, 4, \ldots\}$

an identity element is an element if added to a number and there is no change in the result that additive identity or element is the identity element.

e.g; $0 + 2 = 2$

the additive element doesn't belong to natural number $0 \notin N$.

for example in the case of multiplication

$$e \times a = a \times e = a$$

$$1 * 2 = 2 = 2 \times 1$$

In multiplication operation the multiplicative identity is a part of Natural numbers.

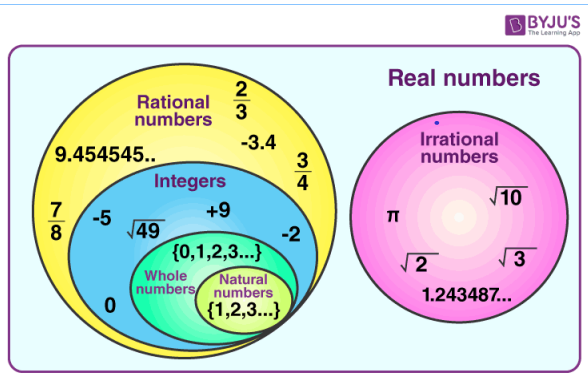
In matrix the additive identity is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

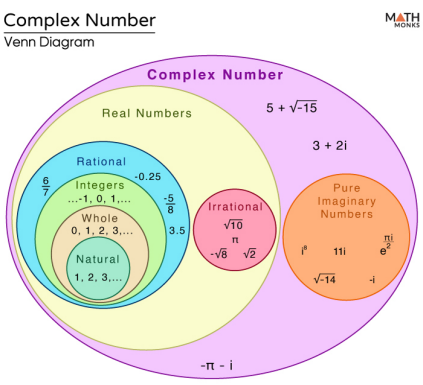
e.g; $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

the multiplicative identity in matrix is

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$N < Z < Q < R < C$.





Group: Definition

An algebraic structure of set G and a binary operation $*$ defined in G i.e. $(G, *)$ is called a group if $*$ satisfies the following postulate:

[G1] Closure: $a \in G, b \in G \Rightarrow a * b \in G$, $\forall a, b \in G$

[G2] Associativity: The composition $*$ is associative in G i.e.

$$(a * b) * c = a * (b * c), \quad \forall a, b, c \in G$$

[G3] Existence of identity: There exist an identity element e in G such that $e * a = a * e = a$, $\forall a \in G$

[G4] Existence of inverse: Each element of G is invertible, ie., for every $a \in G$, there exist a^{-1} in G such that $a * a^{-1} = a^{-1} * a = e$ (identity)

Thus group $(G, *)$ is a monoid in which each of its element is

$$-2 + 2 = 0$$

-2 is the inverse

$$\frac{1}{2} \times 2 = 1$$

$1/2$ is multiplicative inverse.

$(\mathbb{N}, +) \to 0$ doesn't exist due to the additive identity is not inclusive in $\mathbb{N}$.

$(\mathbb{N}, \times) \to 1$ is a monoid since multiplicative identity is inclusive in $\mathbb{N}$.

$2 \in \mathbb{N}$

$\frac{1}{2} \times 2 = 1$

but $\frac{1}{2} \notin \mathbb{N}$ in this case the multiplicative inverse doesn't exist.

$(\mathbb{Z}, +)$ 0

$3 \in \mathbb{Z}$ $\quad \mathbb{Z} = \{\ldots -3, -1, 0, 1, 2 \ldots\}$

$-3 + 3 = 0$

$-3 \in \mathbb{Z}$

hence $(\mathbb{Z}, +)$ 0 is a group.

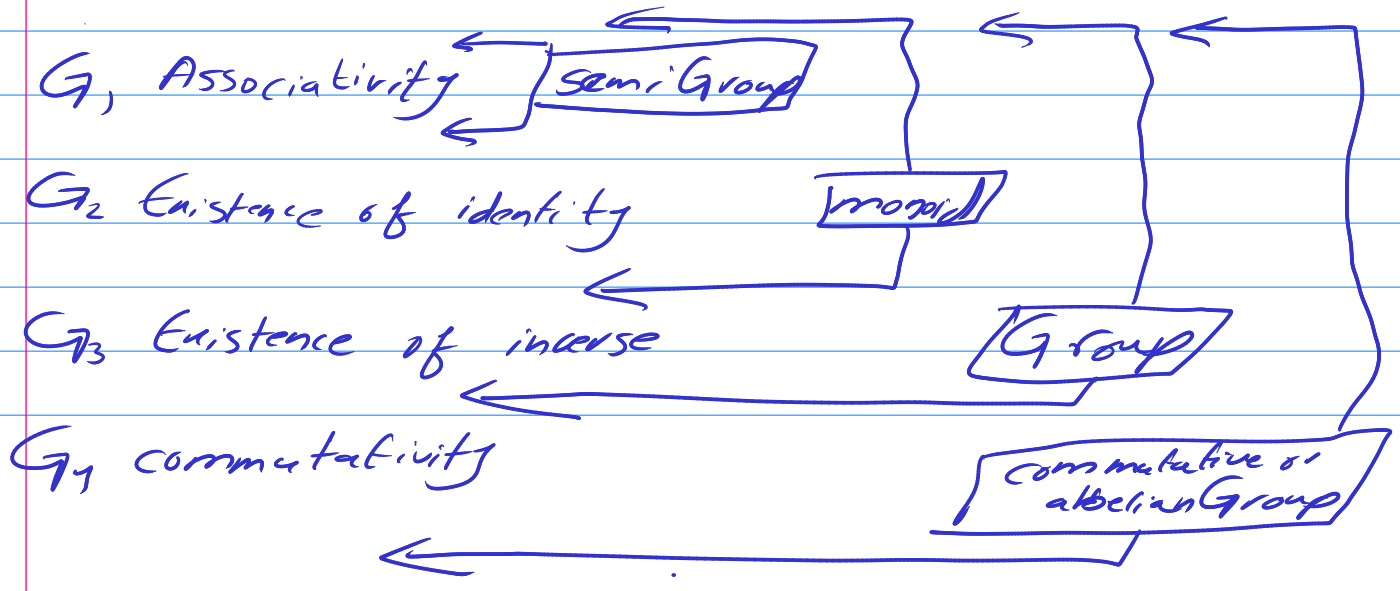
$(\mathbb{Z}, \times)$ 1

$3 \in \mathbb{Z}$

$\frac{1}{3} \times 3 = 1$, $\quad 1/3 \notin \mathbb{Z}$

abelian group or commutative group: Definition
A group $(G, *)$ is said to be abelian or commutative if $*$ is commutative also A group $(G, *)$ is an abelian group, if

$[G_4]$ Commutativity: $a * b = b * a$, $\forall\ a, b = G$



$(\mathbb{Z}, +)$

$a+b = b+a$

$a \times b = b \times a$

$(\mathbb{Z}, \times) \rightarrow$ is not group but

for $(\mathbb{Q}, +)$ it is a group.

hence it is a commutative or abelian group.

finite and infinite Groups:

A group $(G, *)$ is said to be finite if its underlying set G is a finite set

order of a group: Definition

The number of elements in a finite group is called the order of the group.

It is denoted by $O(G)$.

If $(G, *)$ is an infinite Group, then it is said to be of infinite order.